

MARBLE POINT, AntarcticaWMO: **898660**Lat: **77.439S**Lon: **163.754E**Elev: **84**StdP: **100.32**Time Zone: **12.00 (NZL)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-37.1	-35.6	N/A	N/A	N/A	N/A	N/A	N/A	18.5	-19.9	15.4	-21.7	3.2	140	1.189

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.7	3.5	N/A	2.2	N/A	1.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	11.7	9.8	-39.9	6.4	3.0	1.6	-42.1	7.6	-43.8	8.6	-45.5	9.5	-47.6	10.7		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-16.1	-2.3	-7.6	-16.2	-21.2	-21.8	-23.3	-25.3	-25.7	-23.1	-16.7	-7.6	-2.2		
	DBStd	9.83	2.46	3.64	4.29	5.50	6.15	5.68	6.24	6.37	5.58	4.47	3.95	2.50		
	HDD10.0	9533	381	493	811	935	987	999	1094	1106	994	827	529	377		
	HDD18.3	12575	639	727	1069	1185	1246	1249	1353	1364	1244	1086	779	636		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	3.8	3.5	4.0	3.7	3.3	3.7	4.2	3.9	3.9	4.1	3.6	3.9	3.7		
(15) Precipitation	PrecAvg	212	23	24	17	13	16	19	13	19	17	17	17	17		
	PrecMax	469	49	60	69	35	48	62	37	73	58	61	52	62		
	PrecMin	47	2	1	1	5	1	2	0	0	0	0	1	2		
	PrecStd	108	16	17	15	7	13	14	9	18	13	15	11	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	1.6	-5.4	-6.2	-5.4	-6.7	-7.4	-7.7	-7.9	-4.8	1.9	5.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	3.5	-0.1	-7.2	-9.0	-8.6	-10.2	-10.9	-11.2	-10.3	-7.3	0.1	3.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	2.3	-1.5	-8.9	-11.2	-10.9	-12.8	-13.7	-13.5	-12.5	-9.1	-1.2	2.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	1.2	-2.7	-10.3	-13.7	-13.3	-15.0	-16.3	-15.7	-15.0	-10.6	-2.5	1.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	5.0	5.5	6.0	6.3	6.5	7.1	7.1	6.8	5.8	4.7		
		MCDBR	5.7	6.1	6.5	6.8	7.7	8.4	8.0	8.3	8.7	8.2	6.7	6.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.220	0.176	0.112	N/A	N/A	N/A	0.110	0.165	0.196	0.216	0.232		
	taud		2.324	2.336	2.227	1.911	N/A	N/A	N/A	1.894	2.169	2.290	2.322	2.322		
	Ebn at Noon		976	907	781	160	0	0	0	140	778	937	996	999		
	Edh at Noon		91	71	45	4	0	0	0	4	46	74	91	97		
(44) All-Sky Solar Radiation	RadAvg		8.11	4.79	1.68	0.15	0.00	0.00	0.00	0.03	0.91	3.68	7.48	9.36		
	RadStd		0.23	0.16	0.07	0.01	0.00	0.00	0.00	0.00	0.05	0.12	0.20	0.26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page